

A3 — Software Tools and Documentation

Semi-Automatic Bottle Labeling Machine | Team A17 | 2026

Overview

This document outlines the software tools used during the development of the semi-automatic bottle labeling machine. The project was hardware-centric in nature, and the software stack was kept minimal and appropriate to the scope of the system.

Arduino IDE

The primary software tool used in this project was the Arduino IDE (Integrated Development Environment), used to write, compile, and upload firmware to the Arduino Nano microcontroller.

The firmware was written in C++ and handled all core machine logic including stepper motor pulse generation, rotary encoder input reading, I2C LCD display output, and the state machine that governs the labeling cycle.

The following libraries were used within the Arduino IDE:

- Wire.h — for I2C communication between the Arduino Nano and the LCD backpack (PCF8574T).
- LiquidCrystal_I2C.h — for controlling the 16x2 LCD display over the I2C bus.
- KY040.h — for reading the rotational direction and button state of the HW-040 rotary encoder module.

HMI (Human Machine Interface)

This project does not include a dedicated HMI software application. The decision was made based on the standalone, embedded nature of the machine. The system was designed to operate entirely without a connected computer, making a software-based HMI unnecessary and outside the intended project scope.

User interaction is handled directly through the onboard hardware interface: the rotary encoder knob for parameter input and the 16x2 LCD display for real-time feedback. This embedded interface is sufficient for the operational requirements of a semi-automatic labeling workflow in a small-scale production environment.